Revit 3D Modeling Optimizes Substation Design

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***Abstract*— Designing substations and generating drawing packages using 2D software is a mainstay in the power industry. However, the engineering services market is pushing companies to explore options for 3D modeling software to remain competitive. 3D design has the possibility to drive a more integrated process and eliminate inefficiencies that arise during manual 2D drafting. This method could streamline design changes and communication across disciplines, cut down on drafting workload, and change how engineers have approach even boilerplate designs for substations.**

This paper reviews a 3D substation pilot project developed using Revit software, a software widely used in architecture that was adapted for engineering purposes. Instead of creating a drawing package, this software focuses on creating a design from which individual components can be broken out. The benefits, lessons learned, and potential industry-wide impact of adopting 3D software for substation design will be further explored in this paper.

Keywords—Three-Dimensional (3D), Two-Dimensional (2D), Building Information Modeling, Revit

I. 2D DRAFTING

Currently, many service providers in the power industry use 2D Computer-Aided Design (CAD) drafting solutions to design substations. 2D CAD generates graphical, object level displays of information within sets of drawings that capture and then present information about the design as a series of drawings [1]. With this method design notes and calculations must be manually added to the drawings, and software used to run calculations is often housed within an entirely different application. Additionally, any changes to 2D drawings must be marked up either on hardcopy printouts or via PDF editing software.

One of the most common sources of error with the 2D method is that markups to a drawing set are missed or misinterpreted. Additional engineering time is used to verify that markups have been correctly incorporated. This process becomes even more cumbersome on brownfield sites, where multiple sets of drawings that show additions and removals to the substation must be developed and tracked.

For consulting firms, substation design is an area that historically has the highest rate of project write-offs due to inefficiencies and miscoordination. Ulteig began exploring a 3D initiative in 2020 to address these engineering pain points and expand their service offerings to clients.

II. CHOOSING A 3D MODELING SOFTWARE

When determining which software to adopt for the pilot project, the team started by reviewing client requirements. As was the case with this pilot, implementing a 3D service offering may ultimately require a company to invest in multiple software systems. Both Bentley and Autodesk were identified by the team as companies with software applications to be explored in order to create a robust range of 3D services. Ultimately, Revit was chosen as the software to create the pilot project. Revit allows multiple resources to be in the same file at the same time and allows for multiple discipline environments, so collaboration is easier than with other applications. To get a better baseline of project costs, efficiencies, and engineering and drafting time, an existing 2D substation design was used to go through the 3D design and modeling process.

III. BENEFITS OF MODELING IN 3D

Building Information Modeling (BIM), a subprocess of 3D, is a nongraphical software modeling solution that guides design and construction. BIM has three characteristics [1]:

(1) Models are created and operated on digital databases for collaboration.

(2) Models manage change throughout the databases so that any changes are updated throughout the databases.

(3) Information is captured and preserved for re-use in other applications.

Database-based storage is a key advantage that BIM solutions have over 2D modeling. Changes in data that occur over the lifecycle of a project are captured and propagated automatically instead of manually [1]. These models allow information regarding design intent to be built into the model as it is developed, thus keeping any notes on the design in one place [1]. BIM also allows the user to interact with the model, providing contextual information that cannot be shown in 2D [2].

The advantages of a BIM application were confirmed over the course of the pilot project. The engineers and drafting technicians noted that centralization and smart automation of design, cross-discipline collaboration, automatic generation of drawings, and potential for reduction of costs were all significant. Thus far, the project has confirmed that the 2D approach to design is more time consuming, inconsistent, and inherently inaccurate than designing in 3D. These findings will be explored with analysis of their impact on specific portions of the substation design process in subsequent sections of this paper.

A. Document Control

Document control is an important component of any project. For 2D projects, clients often request drawing indexes, and at later the stages of a design, annotations of changes. The Revit 3D model cuts down on the need for these administrative tasks because it stores all files related to the design within one centralized file. The model itself is housed within one database, and all aspects of modeling are

978-1-6654-8167-0/22/$31.00 ©2022 IEEE
DOI 10.1109/REPEC55671.2022.00023

completed within the Revit program. Each component of the design has bi-directional association within the database, meaning that when one aspect of the design is changed, that change is implemented across the entire model. Designers from physical, structural, and protection and controls disciplines are all working within the same model file, so interdisciplinary collaboration is built into the process. Drawing sets for each discipline are generated and printed from the model database itself, and the model contains the aggregate of work done on the project. Because every discipline is working from the same location, 3D modeling also has considerable potential to standardize designs and develop documentation. Drawing indexes and notes on design changes can all be housed within the model database.

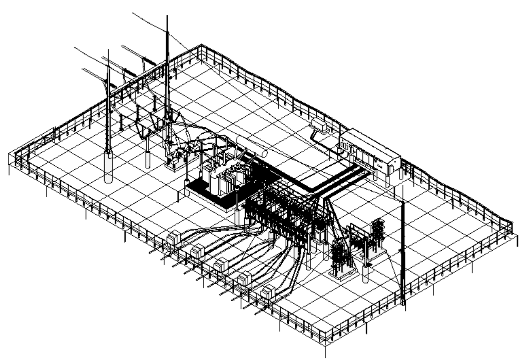

Fig. 1. 3D Image of Pilot Substation

With Revit, project templates and content libraries for individual clients based on those client standards can be created. The 3D team anticipates developing different templates for different clients over the course of Revit integration throughout the company.

B. Cost Reduction

Designing in 3D is much more cost-efficient than designing in 2D. Building information modeling streamlines the process, allowing fewer people to work with less possibility for mistakes. For the 2D process, a design change is implemented by one discipline. That change must be communicated with other designers and follow-up is needed to ensure that other portions of the design are updated. This system lends itself to many inefficiencies (as well as potential errors). With the centralized model, a change by one discipline automatically updates the entire project, thus reducing the need for the additional engineering hours to implement design changes. Reduced human error equals reduced engineering costs and translates directly to reduced project costs. The centralization of the model and bi-directional association of the design also mean that fewer people can accomplish more.

Typical milestones for substation designs are the ten percent, thirty percent, sixty percent, ninety percent, and construction packages. Projects typically kickoff at the ten percent or thirty percent stage. The Ulteig pilot project started the design at the thirty percent milestone. One of the findings was that a BIM thirty percent design package will be much more built out than its 2D counterparts. A 2D thirty percent package normally contains only a General Arrangement, One-Lines, and information on the Control Enclosure and panels. Due to the interconnectivity of the model, the BIM thirty percent package required designers to model the entire substation to approximately the ninety percent stage – so a full site was built out with all equipment and jumper and bus connections. This upfront investment of time and resources allows for better visualization and comprehension of all aspects of the design by all parties. Additionally, changing a project in preliminary stage is much less expensive than doing so at a later stage. A change at the ten or thirty percent stage is much simpler to implement than a change at a sixty or ninety percent stage, and less likely to result in a change order to a client.

C. Design Quality

Linking multidisciplinary items together within the model led to immediate gains in design quality. As an example, a high side circuit breaker would be added to the model by the physical discipline. This breaker has the ability to link to the breaker in the Protection and Control Drawing package. If the breaker equipment number or quantity were to change, the change would be updated across the board on all diagrams connected to the model and the model itself.

D. Applications

The Revit model was found to have different strengths and limitations across different deliverables. The greatest gains were in the physical design realm. This section will highlight specific aspects of a substation design and findings of the pilot team.

The ability to automatically schedule a full Bill of Materials (BOM) and generate specifications was one of the most impactful improvements noted by the physical team. Each piece of equipment in the model is "smart" and had data linked or tied to it, with the part number and equipment-specific information. The program counted the number of times that each part was used in the substation and used the associated information to schedule a BOM and create equipment specifications.

When designing steel, Revit has a functionality wherein each steel member has been built with weights associated to it that will automatically calculate the total weight based on the size of the member. Bus work is another example: When a piece of bus is modeled, bus lengths are automatically calculated and scheduled to the BOM. A typical bus support is taken from a library of parts and then modified to reflect the intended design. This means a baseplate or top plan bolt circle may change to match the equipment required, but the height, steel type, fabrication details and overall design remain unchanged.

Civil grading was performed in Civil 3D, and grading designs were brought into the Revit file. Civil 3D and Revit can be linked back and forth and geo-located. Civil 3D's graded topography can link directly into Revit, and the Revit model can link in Civil 3D for better underground coordination.

Protection and Controls (P&C) engineering was brought into Revit's 2D drafting environment. The out-of-the-box tools that Revit offers to update multiple drawings at once and schedule bill of materials for various drawings such as marshaling cabinets, nameplates, and junction boxes were utilized. The model was useful for visualizing the control enclosure and showing the building footprint, door swing, stairs, piers, cable entry systems, and location of field termination cabinets and relay panels. Coordinating these items with the control enclosure vendor is a pain point and one of the areas of the design that can easily cause delays in the construction schedule. Having a visual, interactive model made it easier for the P&C engineers and designers to check required safety clearances, height of relays and test switches, and ensure that all equipment fit in the enclosure as intended. Additionally, external equipment wiring diagrams can be produced based on vendor information, once that is added to the model. Standardized wiring diagrams for commonly used types of equipment can be generated and tweaked on a project-by-project basis. The pilot team has created templates for common relay vendors so that relay panel design can be partially automated. Automatic generation of AC and DC panel layouts is also another functionality that the team is exploring and expects to achieve.

Grounding and conduit plans and schedule are a deliverable that pulls from P&C and physical resources. The Revit model was able to show underground cable in the substation yard and juxtapose grounding locations with substation topographic information.

The Revit model was especially valuable for permitting activities and understandings of scope. 3D design allows clients and engineers to visualize their design in a way that they cannot in 2D and makes the impact of the design more obvious. 3D designs are also more interactive, allowing different scenarios to be played out. Additionally, visualizing a substation before construction can help with sites where aesthetics is a concern [3].

After some deliberation, lighting studies were not included in the model. While Revit has the potential to model lighting studies, efficient processes and software already exist, and these studies were deemed to be a low-impact item.

IV. LESSONS LEARNED

For all its potential, there are gaps in BIM solutions to address common substation design pain points.

P&C engineering is an area within Revit that required more upfront time for development. One- and three-line drawings, as well as DC schematics are in a 2D environment. Ulteig's solution in Revit is to use the program for its 2D drafting capability as well; this does allow for some connectivity to minimize error between disciplines.

The BIM model greatly reduces the capability for human error, but it cannot eliminate it. Vendor information still must be hand-typed into the model, and any updates to vendor drawings must be carefully implemented and checked to ensure accuracy. However, it should be noted that this capacity for error is also present, and to a greater degree, with traditional 2D drafting systems.

Developing a 3D service offering comes with high front-end expenses for engineering firms. Before executing a pilot project, a sizable time investment was required to develop the framework to use Revit for substation modeling. Training personnel to work within the program is costly and time-consuming. Trainings for multiple types of software may be necessary depending on a firm's clients and their needs, and likewise developing templates within several types of software is also expensive and requires ongoing maintenance and updates. In the power industry, one size certainly does not fit all. Firms may be able to negotiate with clients who are willing to update their own standards, but they must come to the table with concrete data on why such a change is worth it to the client. During the pilot the team experienced pushback from clients who were uncomfortable with the idea of developing new standards.

V. INDUSTRY IMPACT

Despite its flaws, BIM modeling has enormous potential to improve how service providers approach design engineering. The 3D approach to designing substations is reforming the engineering services market, allowing for efficiency and automation to coincide with innovation.

Fig. 2 Rendering of Pilot Substation

BIM models are also segueing into virtual and augmented reality applications. Some of these applications may eventually allow personnel to view a built-out substation model from their mobile phones [3]. This application could reduce overhead costs for maintenance and repair. Access to a visualization of a site allows personnel to train, familiarize themselves with a substation, and address some of the problems that occur on sites without ever having to take the time to travel to the site [3]. Utilities in particular are sensitive to the resources that must be deployed to maintain and repair infrastructure.

On the design side, the pilot project accelerated and streamlined the design process, leading to overall project cost reductions, and greater efficiency, collaboration, and automatization. The exact time savings have not been quantified, but other BIM initiatives in the power industry have reported savings of up to 70 percent for greenfield sites and 50 percent for brownfield sites [3]. Thus far, the team's experience with this pilot project corroborates these findings. The ability to view every aspect of a substation before it has been built creates incredible value for every entity involved in the design process. Given these advantages, 3D modeling is likely the future of the engineering services industry.

ACKNOWLEDGMENT

The author thanks 3D Development and Pilot Project members Dane Swartz, Melissa Wastell, Thomas Bilotta, and Michael Severin for their assistance in researching and developing this paper.

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